INTERACTIVE LEARNING TOOL FOR DYSLEXIA KIDS



RP26-IT-130

SLIIT
FACULTY OF COMPUTING

WHAT IS DYSLEXIA ?

Dyslexia is a relatively common, lifelong learning difficulty affecting up to 1 in 10 people. It mainly causes problems with reading, writing and spelling. It does not impact intelligence.



MAIN WEAKNESSES

Poor Working memory

Phonological awareness

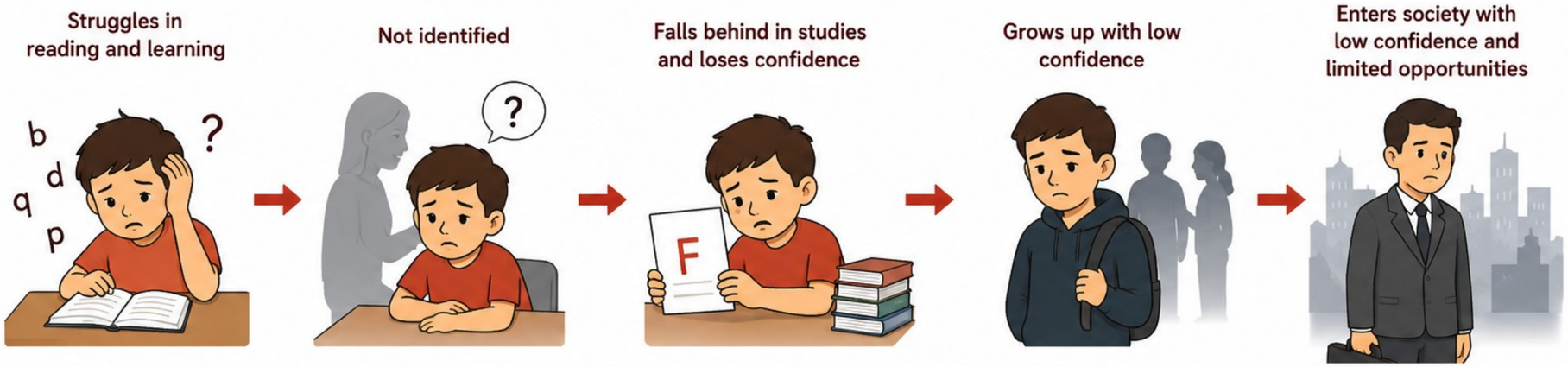


Reading Difficulty

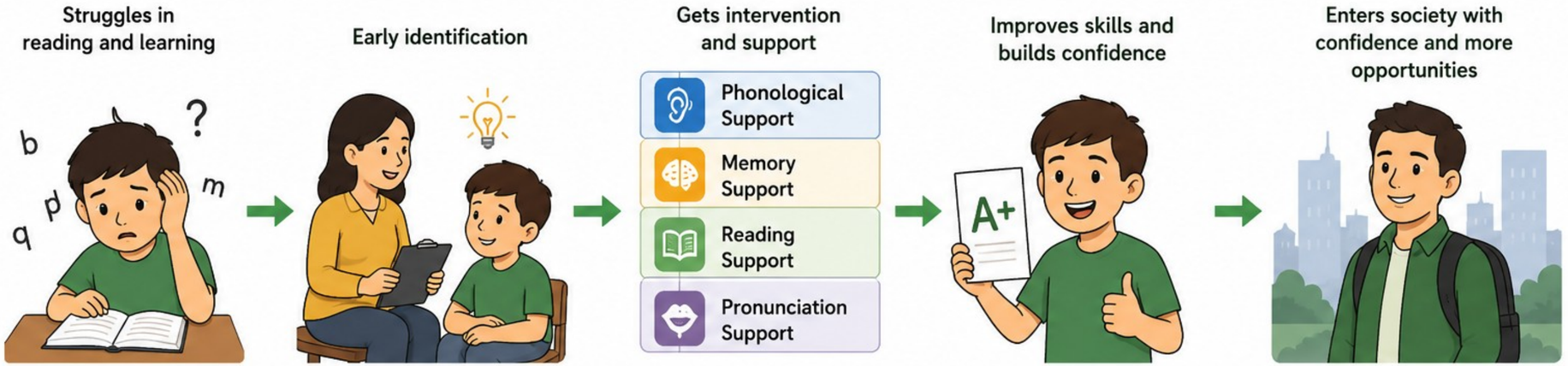
Pronounciation Diffyculty

IMPORTANCE OF EARLY IDENTIFICATION

WITHOUT IDENTIFYING DYSLEXIA



WITH EARLY IDENTIFICATION AND INTERVENTION SUPPORT



TARGET GROUPS



Students



Teachers



Parents

ANY EXISTING SOLUTIONS ?

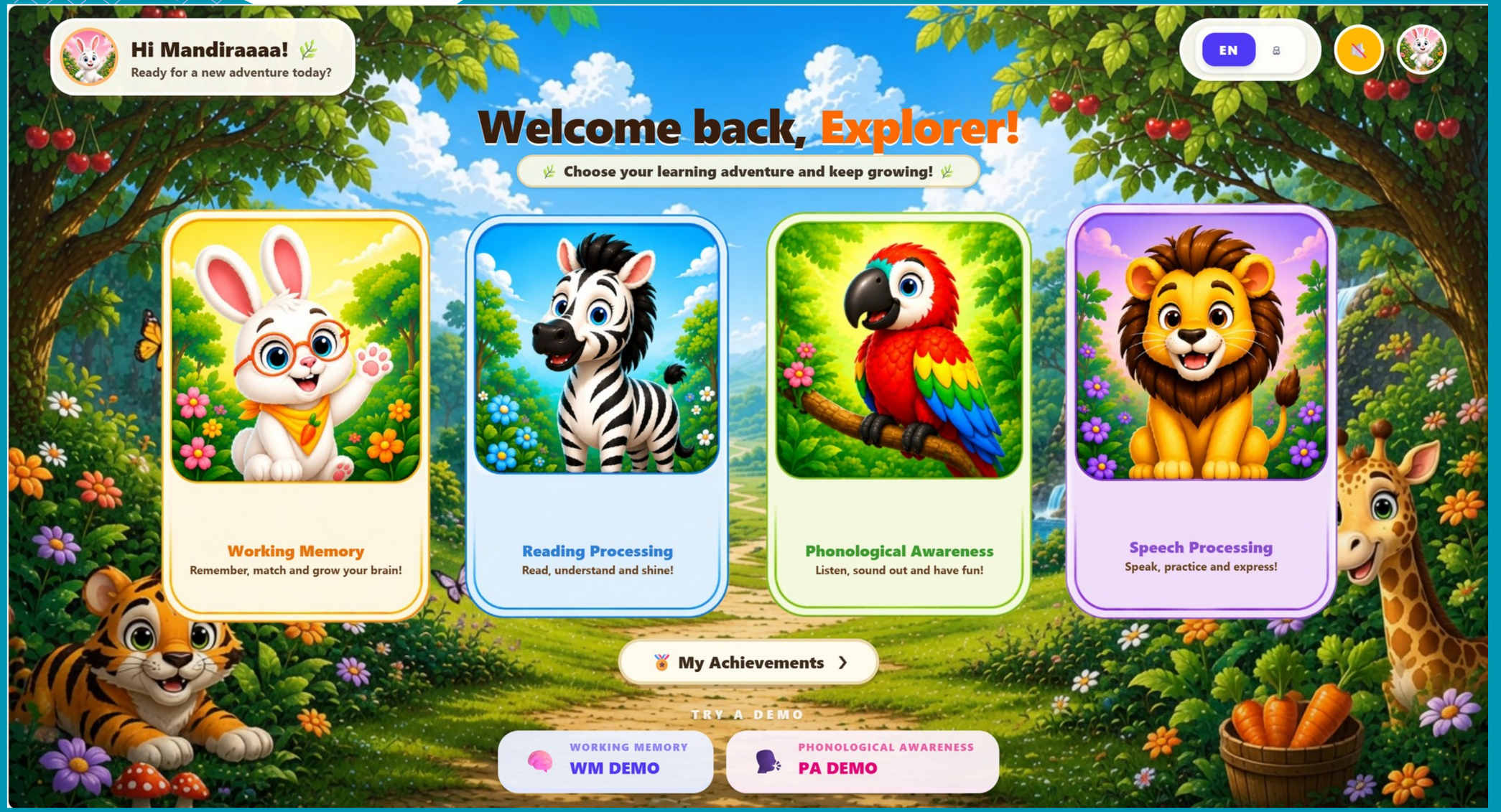
- None of them address all 4 weakness together in a one single platform
- None of them focus on flexibilty for grade 2,3,4 students (Specially based on Local Syllabus)
- Language flexibility between Sinhala and English language (English for international school students)

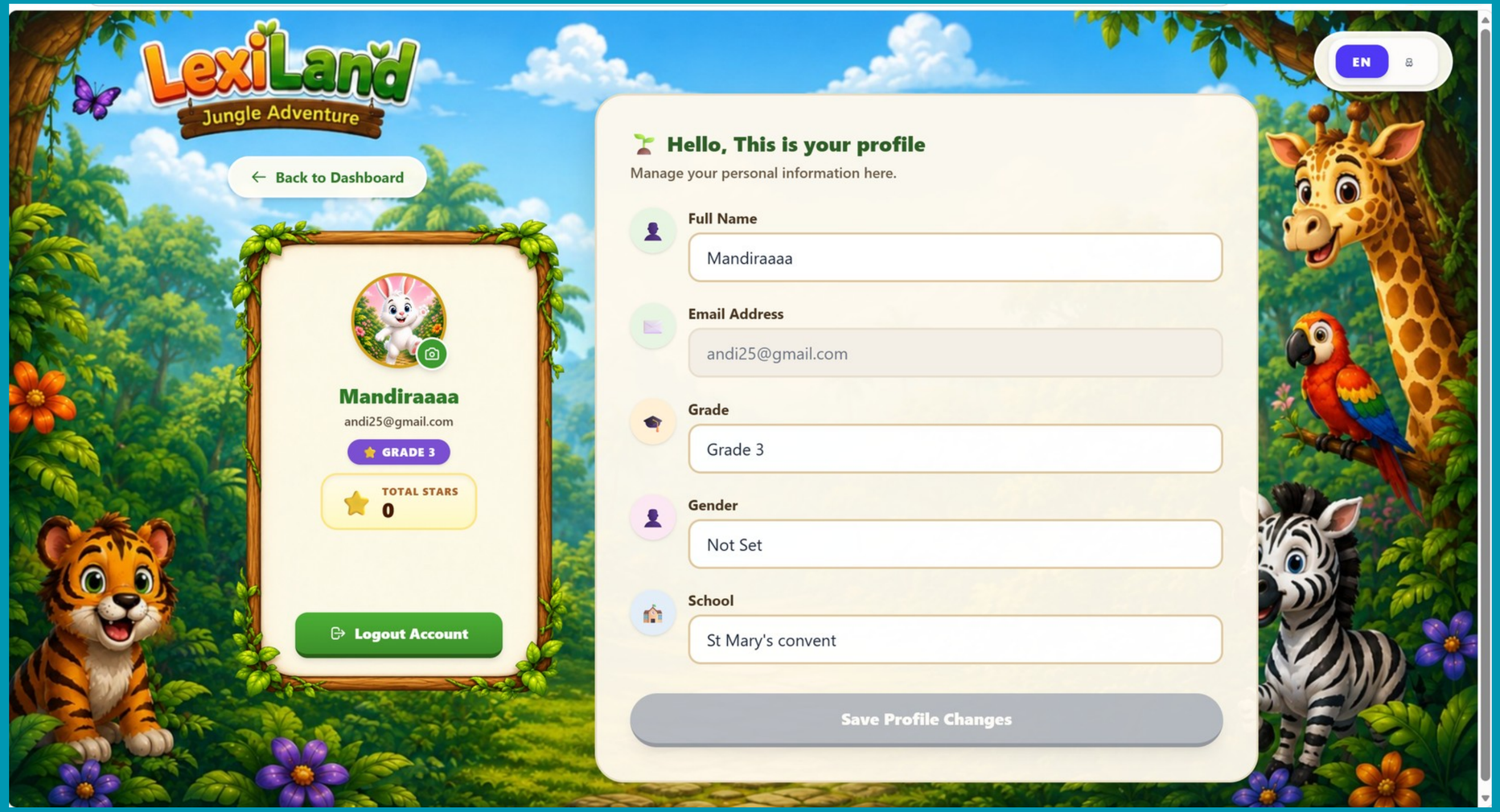
OUR OBJECTIVES

- Build one integrated platform covering working memory, phonological awareness, reading and pronunciation together.
- Align content to the Grade 2-4 local Sri Lankan syllabus.
- Support both Sinhala and English, so it works across local and international schools.
- Screen Test and then categorize learners early using ML, then personalize the practice path.

SOLUTION


LexiLand
Jungle Adventure
Welcome Back
Email Address
andi25@gmail.com
Password
Forgot password?
••••
Sign In
Don't have an account? Create an account


Hi Mandiraaaa!
Ready for a new adventure today?
EN
Welcome back, Explorer!
Choose your learning adventure and keep growing!
Working Memory
Remember, match and grow your brain!
Reading Processing
Read, understand and shine!
Phonological Awareness
Listen, sound out and have fun!
Speech Processing
Speak, practice and express!
My Achievements
TRY A DEMO
WORKING MEMORY
WM DEMO
PHONOLOGICAL AWARENESS
PA DEMO


LexiLand
Jungle Adventure
EN
Back to Dashboard
Mandiraaaa
andi25@gmail.com
GRADE 3
TOTAL STARS
0
Logout Account
Hello, This is your profile
Manage your personal information here.
Full Name
Mandiraaaa
Email Address
andi25@gmail.com
Grade
Grade 3
Gender
Not Set
School
St Mary's convent
Save Profile Changes

NON FUNCTIONAL REQUIRMENTS

- Performance - Lightweight web that loads quickly on shared, low-spec school lab computers validated during on-site testing.
- Security & Privacy - Student results and categorization are access-restricted; no unnecessary personal data is collected from children.
- Usability & Accessibility - Grade-appropriate MCQ / drag-and-drop interactions, full Sinhala + English UI, audio and icon cues for early readers.
- Realiability & Compatibility - Consistent Start→Screen→Score→Level→Progress pipeline across all 4 modules

COMMERCIALIZATION PLAN



THANK YOU !!

APPENDIXES

LexiLand

Commercialization Plan

LexiLand serves two kinds of customers — **schools & learning centers** that license seats for a whole classroom, and **families** who sign up directly for one or more children at home.

For Schools & Learning Centers

Bulk licensing for grades 2–4, managed by a School Admin dashboard.

Monthly Plan

Rs 500 / student / month

Billed per enrolled student, per month

Live

Annual Plan

Rs 4,800 / student / year

≈ 20% cheaper than paying monthly

Live

- Admin dashboard: student profiles & results
- All 4 learning modules, Grades 2–4
- Risk scoring, session history, Excel export
- Contract value auto-calculated by seat count

For Families

Tiered by number of children — same app, home use.

Individual

1 child

Free trial for a single learner

Early access

Plus

Up to 5 children

Small family or micro-practice

Early access

Premium

Up to 100 children

Clinics & private practitioners

Early access

- Guardian-owned student profiles
- Progress tracking per child
- Pricing to be announced — join the trial now

School Admin

Teachers, therapists & parents managing their own students — results, risk levels, and progress in one place.

Super Admin

The LexiLand team — runs schools, users, billing and platform quality across the whole product.

Watch & Learn with LexiLand

We also run a YouTube channel, **LexiLand**, where we post learning videos, activity walkthroughs, and product updates for schools, parents and teachers.

LexiLand | School plans live today · Family plans in early access | YouTube: LexiLand

LexiLand

Start Your Educational Adventure

Choose your learning adventure and keep growing!

Four Pillars of Literacy

Memory, sounds, reading & speech — all in one app.

Assess → Personalize → Practice

Every child gets the level that's right for them.

Fun For Kids, Clear Data For Grown-Ups

Stars for learners, reports for parents & schools.

Four Learning Modules

Working Memory

"Remember, match and grow your brain!"

- Memory Check Adventure finds the right starting level
- Adaptive difficulty, easy → super
- Picture memory, digit span, sequences & patterns
- Earn stars, unlock the next challenge

Phonological Awareness

"Listen, sound out and have fun!"

- Sound Detective, Picture Match, Word Play & more
- Builds hearing, matching & sound-manipulation skills
- Progress charts show skills growing over time
- Supports reading & spelling foundations

Reading Processing

"Read, understand and shine!" — Reading Friend

- Optional eye-tracking gives gentle hints when needed
- Shapes → letters → words → sentences
- Practice mode works even without a camera
- Warm, dyslexia-friendly reading experience

Speech Processing

"Speak, practice and express!" — Leo's Sound Safari

- Leo the Lion guides a jungle sound adventure
- Read aloud, get friendly feedback
- Echo Roar, Robot Words, Sound Twins & more
- Collect badges, gems and stars

Who LexiLand Is For

Children

- Learn through play, not pressure
- Clear path: Identify → Improve
- Stars & badges keep practice fun
- Friendly characters & bilingual (EN/SI)

Parents & Teachers

- Student profiles, all in one place
- Skill levels & risk indicators (WM & PA)
- Session history & Excel export
- Privacy-aware — see only your students

Schools

- All four modules, one platform
- Per-student licensing, any school size
- Grades 2–4, admin dashboard for SEN teams
- Monthly or annual institutional plans

Designed for learning support, not medical diagnosis. Reading gaze data is used for hints and progress only. Speech support is a provisional learning indicator, not a clinical label.

Schools: from Rs 500 / student / month (or Rs 4,800 / student / year) · Individual & family tiers available (1 / 5 / 100 learners)

START YOUR ADVENTURE

Grades 2–4 | English & Sinhala | Home & School | Contact us for school demos & pricing

Learning videos & walkthroughs on our YouTube channel — **LexiLand**

Premium LK

lexiland

Create

Home

Shorts

Subscriptions

You



LexiLand

@LexiLand-m5m • 2 subscribers • 6 videos

This channel provides informative and educational content about dyslexia, including its ...more

Subscribed

Videos



How to Play Pattern

5 views • 4 days ago



Count & Split

2 views • 6 days ago



Word Play

3 views • 6 days ago



Pattern Hunt

19 views • 6 days ago



Picture Match

8 views • 6 days ago



Sound Detective

8 views • 6 days ago

SOUND MINDS ACADEMY

REHABILITATION, EDUCATION, TRAINING

(Gov.Reg.No. W.D.17553)

30/08/2026

The Research Evaluation Panel,
Sri Lanka Institute of Information Technology (SLIIT),
Malabe, Sri Lanka

Confirmation of Child Participation and Guardian Consent for Research Activities

Dear Sir/ Madam,

I am writing to confirm that a child under my care was permitted to participate in age-appropriate game-based learning and assessment activities conducted as part of the research project "LexiLand" being carried out by the student research group R26-IT-130 at SLIIT. Prior to the child's participation, consent was obtained from the guardian for the child to take part in the activities required for research and testing purposes. The activities provided to the child were educational and age-appropriate and included the following:

- Reading activities
- Word and non-word reading activities
- Auditory and sound recognition activities
- Sentence and paragraph reading activities
- Other age-appropriate learning games related to research

The child participated in these activities for the purpose of evaluating and improving the proposed interactive learning system developed by the research group. The activities were conducted in an appropriate manner, with consideration given to the child's age, comfort, and participation.

I further confirm that the research group was informed of the importance of maintaining the privacy and confidentiality of the child. Personal and identifiable information relating to the child was treated confidentially and was not to be disclosed or used for purposes outside the approved research activities.

This letter is issued as a confirmation of the child's participation and the consent and confidentiality measures associated with the research activities.

Yours faithfully,

Beenuka Wickramatunga

Dr. Beenuka Wickramatunga
PhD (Psy), MSc (Clin. Psy), BSc
Psychologist
SRILNAC: 2015/0276 | APA: 00191008

SOUND MINDS ACADEMY, SANDUNPURA, KALAPALUWAWA ROAD, AGGONA, RAJAGIRIYA 0721094505 SMICPSY@GMAIL.COM |

SOUND MINDS ACADEMY
REHABILITATION, EDUCATION, TRAINING
(Gov.Reg.No. W.D.17553)

30/08/2026

The Research Evaluation Panel,
Sri Lanka Institute of Information Technology,
Malabe,
Sri Lanka

Certification of Screening and Improvement Activities for the LexiLand

Dear Sir/ Madam,

I am writing to confirm that I have reviewed the screening and Improvement activities developed by the LexiLand Research Group (Research Group ID: R26-IT-130) as part of their research project for the development of a gamified interactive learning tool for children. Following my review, the 15 screening questions developed for the Working Memory and Phonological Awareness modules were assessed and certified as being suitable for use in the research process to identify children who may demonstrate difficulties associated with dyslexia.

The questions were designed according to the intended age group and the learning and cognitive skills relevant to the respective modules. I have also reviewed the improvement activities developed across the four main learning modules of the proposed system. These activities were designed to be interactive and engaging while supporting the specific learning areas targeted by the research.

The grade-wise improvement activities were further reviewed with consideration given to the Sri Lankan local school curriculum for the relevant grades. The research group developed activities covering the required learning areas in both Sinhala and English, with the intention of providing appropriate learning support for children following either language medium. The questions and activities used within the system were developed and structured by the research group rather than being directly generated by artificial intelligence. The questions were implemented as predefined, hard-coded activities within the system to ensure that the content remained controlled, consistent, and appropriate for the intended users.

Particular attention was given to ensuring that the questions and activities:

- Are appropriate for the intended Grade 2–4 age group.
- Address the relevant skills within the areas.
- Provide interactive learning experiences across all four modules.
- Are structured according to the grade level of the child.

- Cover the relevant areas of the Sri Lankan local curriculum.
- Are available in both Sinhala and English.
- Use child-friendly and appropriately structured questions.

Based on my review, I consider the screening questions and grade-wise improvement activities to be appropriate for the intended research purpose and target age group, subject to the limitations and validation requirements applicable to a research-based screening and educational tool.

This letter is issued to confirm my review and professional assessment of the content developed by the LexiLand Research Group (R26-IT-130) for submission to the Research Evaluation Panel of SLIIT.

Beenuka Wickramatunga

Dr. Beenuka Wickramatunga
PhD (Psy), MSc (Clin. Psy), BSc
Psychologist
SRILNAC: 2015/0276 | APA: 00191008

SOUND MINDS ACADEMY
REHABILITATION, EDUCATION, TRAINING
(Gov.Reg.No. W.D.17553)

30/08/2026

The Research Evaluation Panel,
Sri Lanka Institute of Information Technology,
Malabe,
Sri Lanka

Confirmation Regarding the Suitability of the LexiLand Interactive Learning Tool's UI

Dear Sir/ Madam,

I am writing to confirm my involvement in providing professional guidance and feedback to the LexiLand Research Group (Research Group ID: R26-IT-130), which is engaged in the development of a gamified interactive learning tool for children as part of their academic research project.

The user interface (UI) and overall design of the proposed LexiLand interactive learning tool were presented to me for review. Based on my professional assessment, I found the overall UI design, color palette and level of interactivity to be appropriate and suitable for the intended target age group of Grade 2–4 children.

I provided guidance and assistance to the research group regarding the design and presentation of the tool where required. Particular attention was given to ensuring that the interface was understandable, engaging, visually appropriate and suitable for children within the specified age range.

As part of obtaining practical feedback, the research group also demonstrated the developed tool to children under my supervision. Their observations and feedback regarding the usability, appearance, interaction, and overall experience of the tool were considered by the research group when further improving the system.

The tool presented to me included learning and assessment activities covering several areas relevant to the research, including:

- Working Memory activities
- Phonological Awareness activities
- Reading Process activities
- Speech Process activities
- Interactive and game-based learning activities

Based on my review and the feedback obtained during the evaluation process, I consider the developed interface and interactive approach to be generally appropriate for the intended Grade 2–4 target group and suitable for use as part of the research group's proposed learning and assessment system.

I have provided the necessary professional assistance and feedback to support the research group in developing and refining the tool according to the needs of its intended child users. This letter is issued as a confirmation of my review, guidance, and feedback provided to the LexiLand Research Group (R26-IT-130) in relation to the development of their research tool and validation requirements applicable to a research-based screening and educational tool.

This letter is issued to confirm my review and professional assessment of the content developed by the LexiLand Research Group (R26-IT-130) for submission to the Research Evaluation Panel of SLIIT.

Beenuka Wickramatunga

Dr. Beenuka Wickramatunga
PhD (Psy), MSc (Clin. Psy), BSc
Psychologist
SRILNAC: 2015/0276 | APA: 00191008